Doc V1.5

ESP32 Hardware & Wiring_V4 05/02/2026

This document should be read alongside

Schematic diagram. Note: - Schematics show a bare board, rather than it sitting in the Expansion board.

'Build a Box' document. Details how to mark out, cut and mount everything into a project box.
It also has photos detailing the finished wiring.

All of the boards are pre-made; you will not have to solder the components yourself. Most boards will also have the header pins pre-soldered. The GY-302 (light-meter) and the LED matrix require the header pins soldered. Soldering is also required for the buttons, flash socket and external connectors, if you choose to use them.

JST HX cables are used to connect the different modules to the breakout board. They make a neat solution and fit tightly to the header pins on the modules, so do not work loose like Dupont wires.

There are many 3.3V, 5V & GND power connections. These will not all fit into the breakout board screw-terminals. It will be necessary to connect them in groups with a choc-block or similar, then take one wire to the screw terminal.

Whilst it is not good practice to mix voltages, there is no way to avoid this, as some modules require 3.3V and others 5V. Please take care to ensure the correct voltage is supplied to each module and on the correct pin.

Light-meter, LED matrix and LCD all require 5V. If not using these modules, then 5V wiring is not required.

An optional pcb is available, which replaces the expansion board. They route all of the tFT wiring on the pcb, thus making wiring easier.
They can only be posted within Europe, due to postage costs. See or post on Photrio, for more details.

Ensure you pay particular attention to the polarity of the connections. The boards are clearly marked.

The LCD is marked Vcc and is connected to 5V. (Red wire) and GND, (Black wire)

The GY-302 Light Sensor is marked Vcc and is connected to 5V (Red wire) and GND, (Black wire)

The Laser tx board is marked as S (3.3V) and - (GND) Do not connect anything to the centre pin.
On many Laser boards, the Laser barrel is only held to the board by its wires. This means it can easily be knocked out of alignment. Consider using hot-glue or similar to affix the Laser barrel to the board.

The Sensor (rx) board is marked VCC (3.3V) and GND. The centre pin is marked OUT, which is wired to one of the input pins on the ESP32 Expansion/breakout board.

Be very careful when fitting the receiver sensor into the board. Refer to the photograph. The little lens must be pointing towards the header pins on the board.

Note – The sensor & Laser boards are designed to work from 5v, but normally work fine from 3.3V.
However it has been found that if the wiring is long or thin, the sensors may not trigger correctly.
In which case they may safely be wired to the 5v supply.

The TFT is marked VDD (3.3V) and GND (0V). The TFT board LED pin is also connected to 3.3V

JST HX connectors.

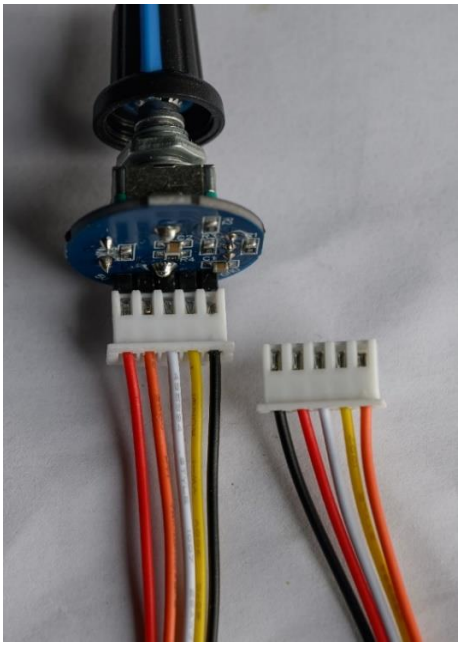
Using the kit containing separate plastic shells and pre-made wires is the easiest option. The wires can then be assembled in the correct colour order.

Note: - The kit does not include Brown wires, or an 8-way connector.

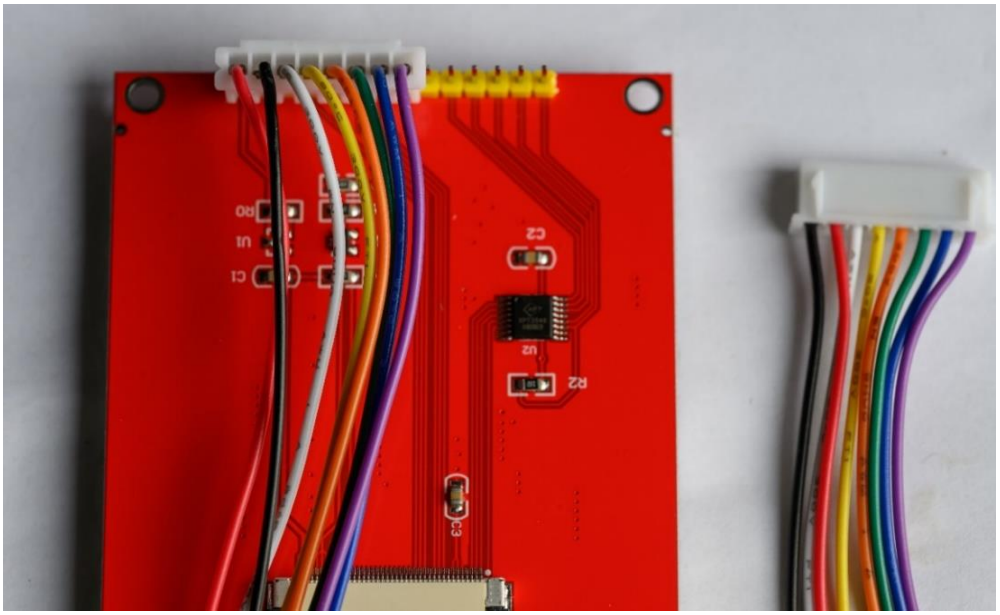
The cheapest solution is to use Red for both 3V and 5V power, but BE VERY careful that the TFT is connected only to 3V.

Update -Touch functionality has been added, so two 7-way connectors from the kit can be used to make a 14-way connector. The plastic lip on the connectors will need to be trimmed so they will sit side-by-side. An alternate is to purchase a separate 8-way connector and a 5-way connector from the kit. They then sit either side of the unused pin at position 9.

Important Note: - wiring colours have slightly changed since these photographs were taken.
Where you see an Orange wire, use Blue instead.



This is the encoder, showing the correct wiring colour sequence.
NOTE the Orange wire shown here. Use Blue instead.
(Ignore the pre-wired connector to the right).



Connection to the TFT screen using an 8-way connector.(Ignore The pre-wired JST HX connector to the right).
The next pin is not used and the remaining pins are for touch and are wired as pre the schematic.

As described above, you can use an 8-way connector (purchased seperatley) or two 7-way connectors from the kit.
If doing this, the sides of the connectors, where they touch, will need trimming.

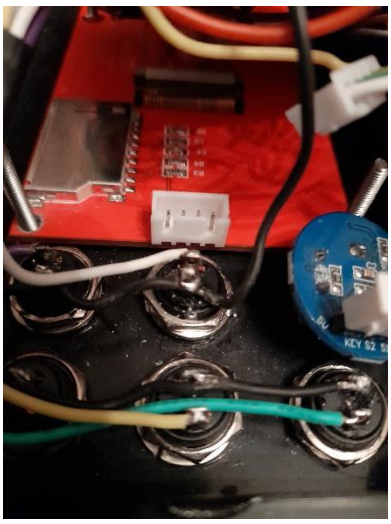
Wiring for the TFT screen, encoder and buttons.
Orange terminal connectors have been glued to the board using epoxy resin.
Top terminal connectors are +3.3V and 0V

Bottom terminal connectors are not yet connected, but will be 0V and +5V

SD card connections.

The SD card slot on the TFT board can be used for data logging if required. SD card must be formatted FAT32 and be no larger than 32Gb

The tft board does not have a connector fitted for connection to the SD card holder, so a 4-way JST HX socket was soldered onto the board and a slot cut into the project box for insertion of the SD card.



LCD.

The LCD is a legacy from the Arduino unit and cannot display as much information as the TFT or PC screen.
The LCD is depreciated and may not be supported in future firmware updates.

In place of the LCD, a larger font and simplified display have been added to the existing TFT screen, allowing the user the option of detailed or larger simplified display.

Note: - The specified project box is too small to house the LCD.

Flash connections.

The flash connection can be used to test when the flash will fire during the shutter cycle, to ensure it is triggered at the correct time. A suitable cable and 3.5mm socket are detailed in the parts document. Two wires require soldering to the socket.

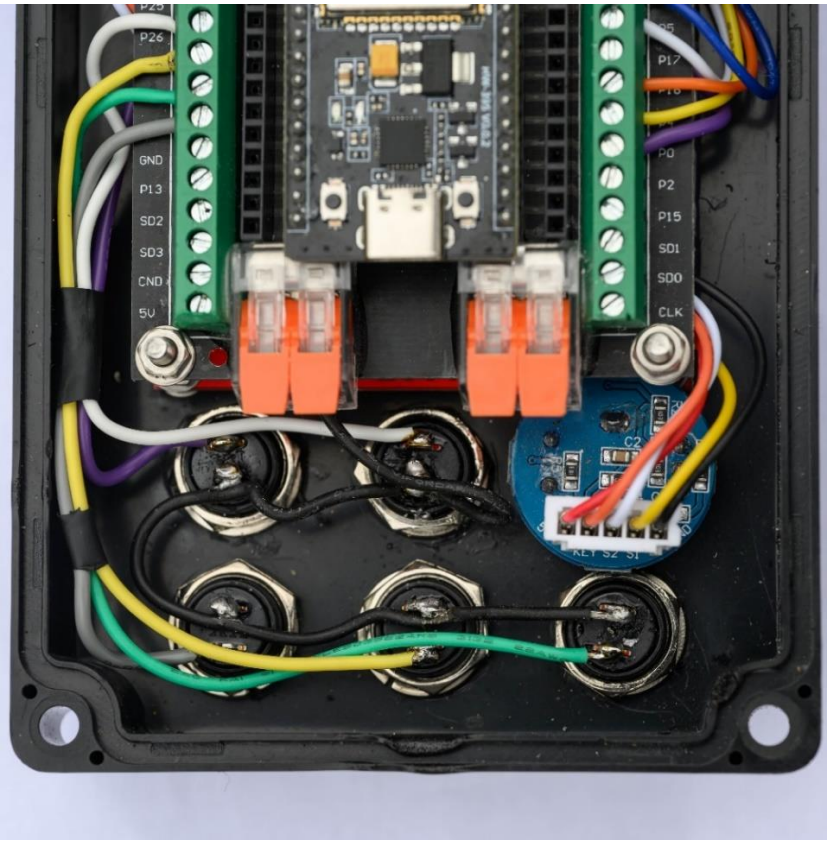
Note: - only cameras with mechanical flash contacts should be connected.

Connections to 3.5MM jack socket.

Button Wiring.

A black wire is connected to one leg on each button in a daisy-chain and then run to the 0V connector block.
The other leg on each button is connected to the corresponding screw terminal on the breakout board.

The wires used were the same colour as the buttons, except for the Black button, where a grey wire was used and the red button, where a purple wire was used, to ensure they were not confused with +v and 0v



Close up of button wiring. Black wire is common to all buttons and then goes to 0V
Grey wire has been used for the Black button and a purple wire for the Red button to avoid confusion with +V and 0V wires.

Other Module wiring

To connect to the external modules, there are a few choices.

a) use JST XH connectors. These are quite short, so will require additional wires soldered onto them.

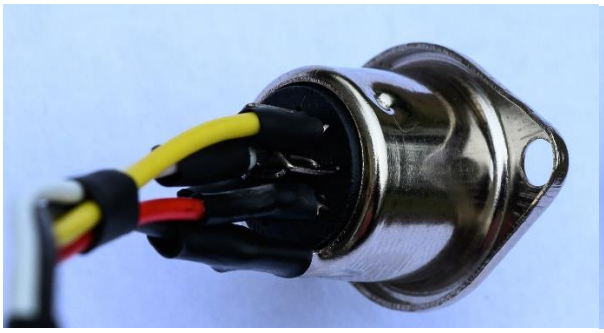
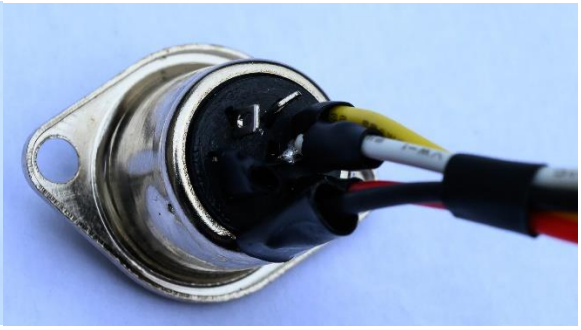
b) The neatest solution is to purchase JST HX bare pin connectors (female type) and a crimp tool, to attach them to the end of the multicore cable. Suitable connectors & crimp tools are available on AliExpress. Or solder the wires directly to the module.

c) Solder the multicore cable directly to the module. This is probably the most suitable option.

To connect the other end of the cable to the expansion board, the multicore cable can either be run directly into the project box through a suitable hole and terminated on the Expansion board (most straightforward method), or connectors used.

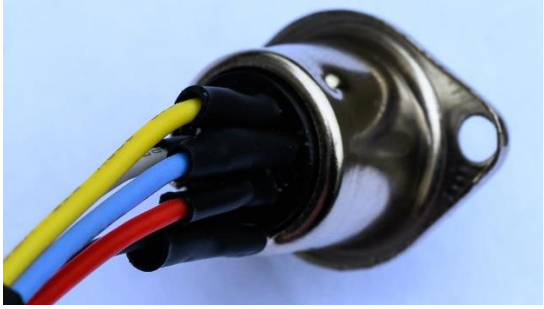
The below details the use of sockets: -

Extreme care must be used if using plugs & sockets, that dedicated pins are used for 3.3V and 5V.
Connecting voltage to a data pin of a module is likely to destroy it.

Wiring for the Light-meter & LCD.
0V bottom large flat pin
+5V is on the bottom right pin
Going anti-clockwise, the next pin is not used,
the next pin is Yellow.
The centre pin is White.

Additionally, the bottom left pin (unused here) has been insulated as it is used for 3V for other sensors.



Wiring for the Light Meter
0V bottom large flat pin+5V Bottom right
and anti-clockwise, the next pin for Blue
Next pin for Yellow.
Centre pin is White

All pins have been insulated with shrink tube, to ensure no short can occur.

Additionally, the bottom left pin (unused here) has been insulated as it is used for 3V for other sensors.

Sensors & Laser wiring.

All pins, marked as 'S' on the Laser boards can be daisy-chained together and connected to +3V
The 0V pins can similarly be daisy-chained and connected to 0V

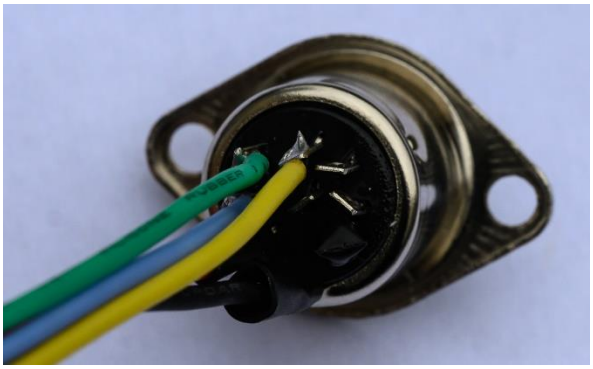
The Vcc pins on the sensor board can daisy-chained together and connected to +3V'
The 0V pins can similarly be daisy-chained and connected to 0V.

The centre pin on each of the sensor boards has its own wire going back to the ESP32 breakout board.
(Yellow, Green or Blue)

The multicore cable can be run directly to the breakout board, or a plug & socket used on the exterior of the project box. Using a plug & socket will allow the sensor frame to be changed, for example from a horizontal to a vertical layout.

If using a plug & socket, an 8-pin version would allow two modules (sensors/Lasers & light meter for example) to both use an 8-pin connector to save money, as they come in packs of 5.

Be very careful with the 3V and 5V wires, ensuring they do not come into contact with any of the data pins.



8 pin Din socket wiring. 0V is the large flat terminal at the bottom.
Red is 3V and used the bottom left pin. (To ensure it is not crossed with the 5V pi used for the other modules)

The pins are very tightly spaced, so the +3V and 0V have been insulated with shrink tube to avoid any short circuits.
Blue, Green, Yellow then use the next three pins, going clockwise.

Additionally, the bottom right pin (unused here) has been insulated as it is used for 5V for other modules.

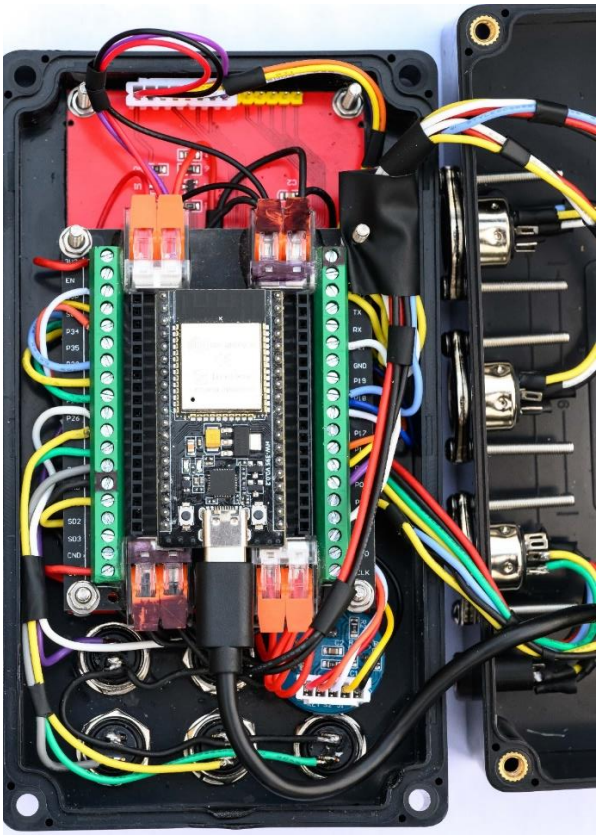
A safer alternate is to use different styles or pin configurations for each of the connectors.
For example purchase
8-pin connectors for the sensor board,
5-pin for the Light-Meter.
6-pin for the LED Matrix (this device is more of a curiosity & most people do not include it)
3.5mm jack socket for the flash.

When making the plug connections. It is best to put the plug into the socket before soldering.
This helps keep the pins straight when soldering and also allows you to match the correct pin colours.
Do NOT forget to put the plastic outer onto the cable first!

Note: - the plugs do not have 0V pin. The 0v wire is soldered to the semi-circular metal cover, near the strain relief clamp.

Completed Unit.

Note Touch wiring on the tft not shown here and the use of the older wiring colours (orange) used.



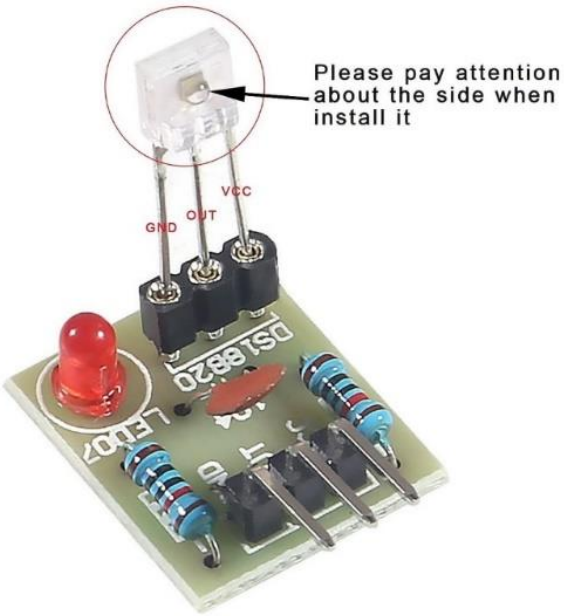
Sensors & Laser mounting

If just testing horizontal shutters, using the block-of-wood frame to hold Lasers & sensors works well. This is an easy way to get up & running. For measuring vertical shutters, the same idea can be used, but obviously the sensors and Lasers must be stacked vertically.

For a finished project, mounting the sensors & Lasers in boxes is the ideal.

O O O

[----------32mm horizontal spacing----------]



Sensor rx – showing correct sensor placement.

Note: - To protect the sensors from damage, they can be omitted before the initial power-up. The LED on the module should light, proving connection & polarity is correct. After de-powering, the sensor can then be fitted to the module.

The project is powered via the USB connector on the ESP32 board. No other power is required. **Before connecting the USB power lead, check, double check and then treble check** all of your connections and the correct orientation of the receiver sensors.

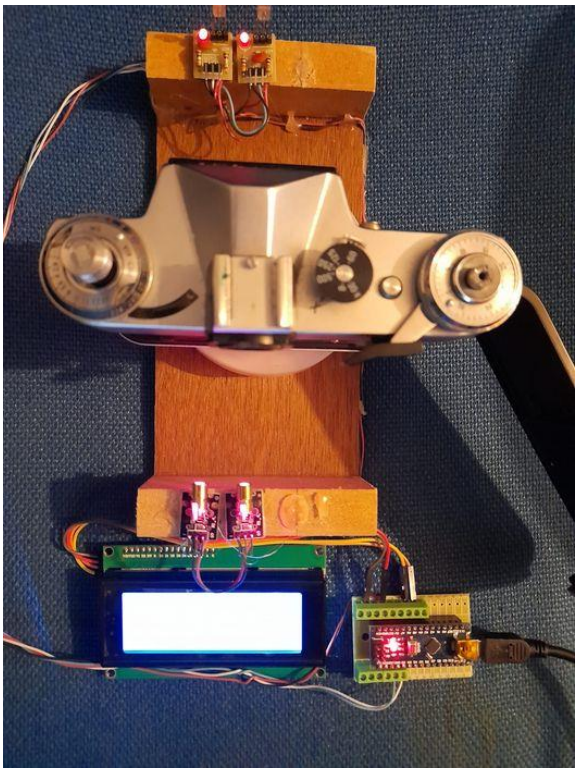
When applying power, the Lasers will shine red and also the red LEDs on each of the rx boards should light.

For the prototype, to make the sensor unit, two pieces of wood were cut, one 40mm high, to mount the Lasers (tx). The other 28mm high, to mount the sensors (receivers, rx) Spacers under the camera (scrap wood, beer mats etc) used to raise different camera models to the correct height.

The Laser & sensor modules were hot-glued to the top of the wood. Hot-glue allows the lasers to be moved for alignment, whilst the glue cools.

The below photo shows the Arduino prototype, which used two sensors and Lasers.

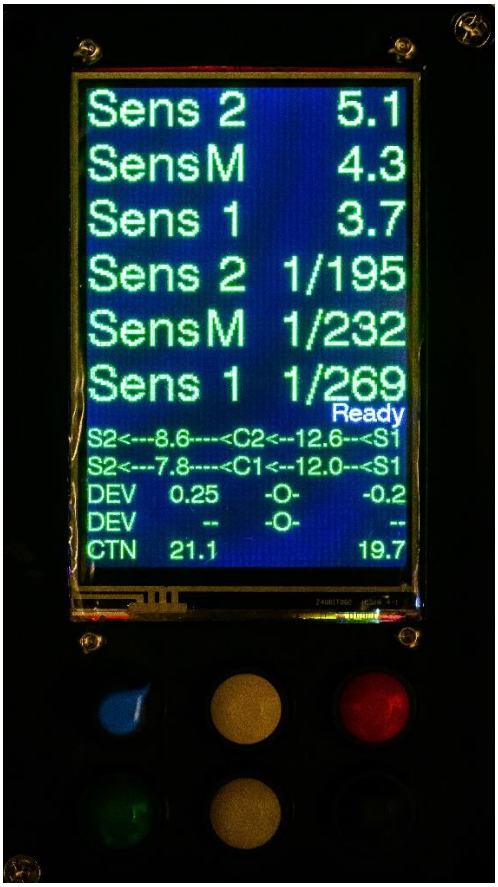
Using this method is a quick simple way to get everything working.
3d printable parts are now available, including a case (rather than using the project box)
and Sensor & Laser boxes.



Prototype shutter tester with two Lasers.

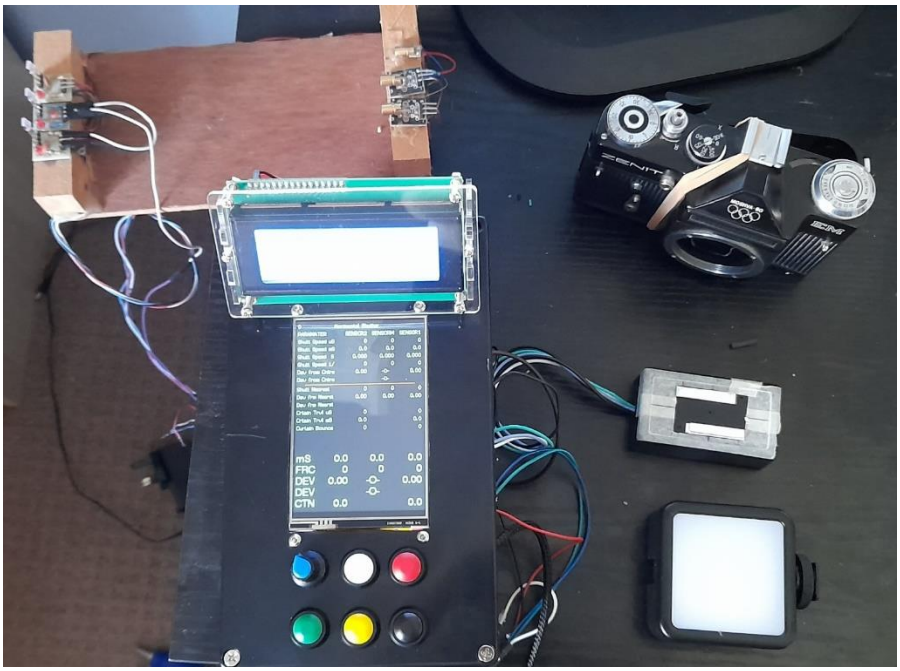
In this example, wires (taken from an Ethernet cable) are directly soldered to the Laser boards, rather than jumper wires being used.

This simple design works perfectly well & the third Laser & sensor for the ESP32 version easily added.



Here is the 158-90-60 project box with just the TFT display.
(It is displaying the Simple Screen, which has been added to replace the LCD).

Here is the ESP32 Shutter Tester in a larger 200-120-75 project box, with both TFT & LCD. The LCD is mounted in an acrylic frame and bolted to the top of the project box.



The original block of wood Laser and sensor frame can be seen at the top, now with three sensors and Lasers.

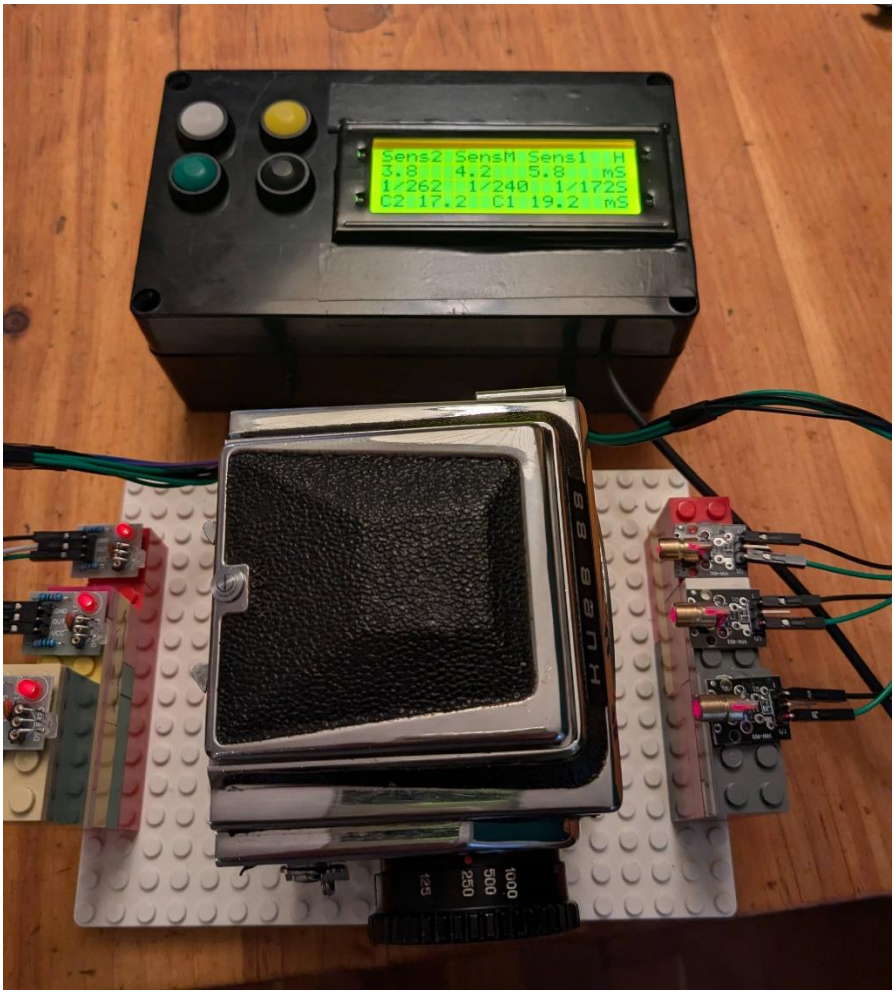
Below the camera is the film-gate sensor and LED light box used instead of the Lasers.

Note: - Lasers work far better than a LED light source and film-gate sensor. Some people like experimenting with the 'traditional' way of using a film-gate-sensor, thus details of how to make one are detailed in a separate document.

Here are some pictures from the Photrio thread, showing some of the designs people have made, many are the Arduino versions.

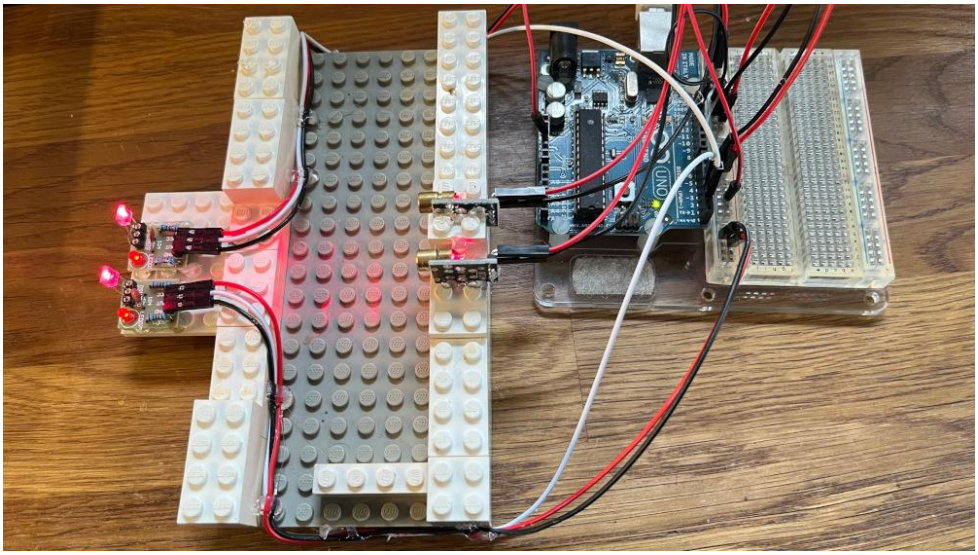
Compact version in 3d printed case

Inside the compact version.



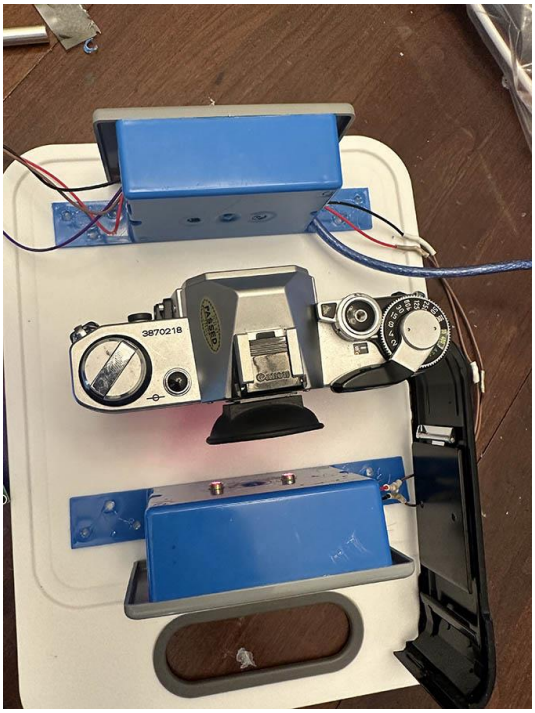
Lego version, testing a medium format camera.

3d printed version. Note lasers & sensors are arranged diagonally, 32mm spacing horizontally & 20mm vertically.
This way the same sensor frame can be used for horizontal or vertical shutters.
It does however make it a little trickier to line up the film-gate with the Lasers.

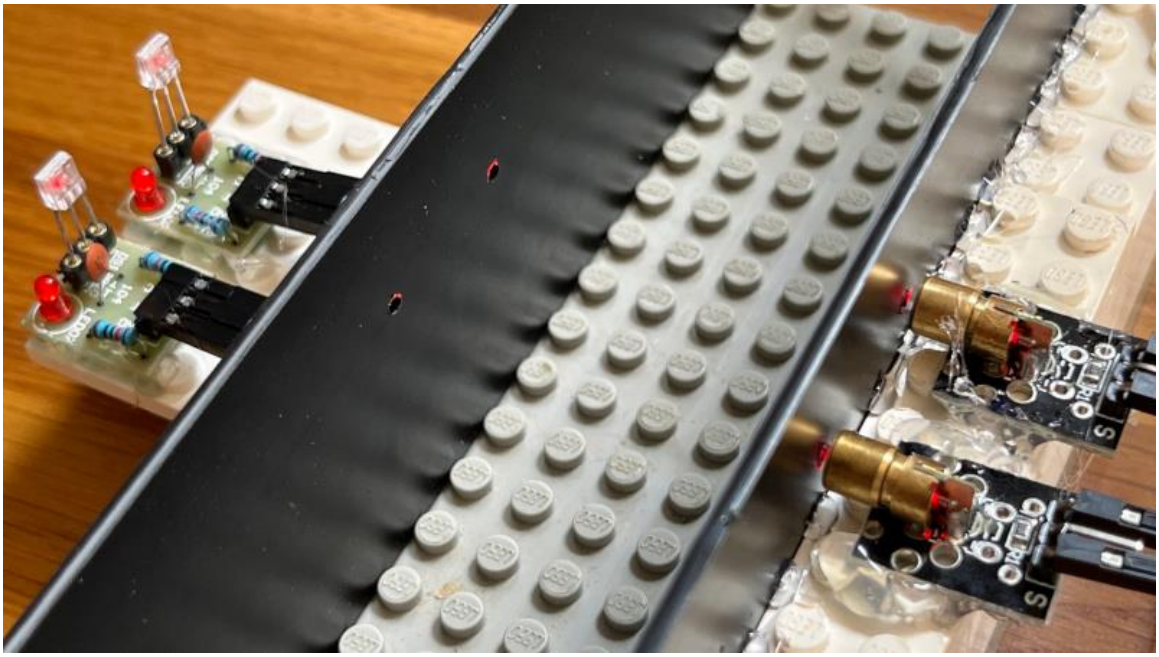


Lego :o)

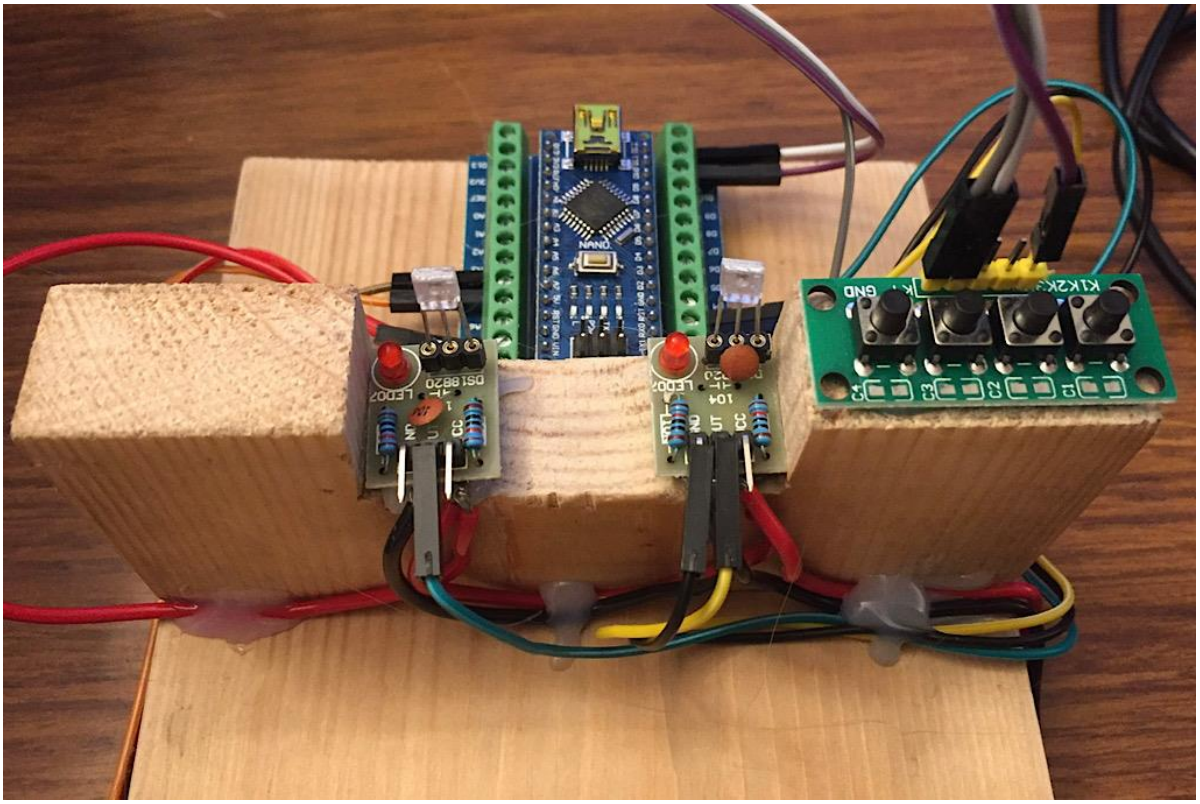
Horizontal or vertical shutters. Switch toggles horizontal or vertical Lasers.



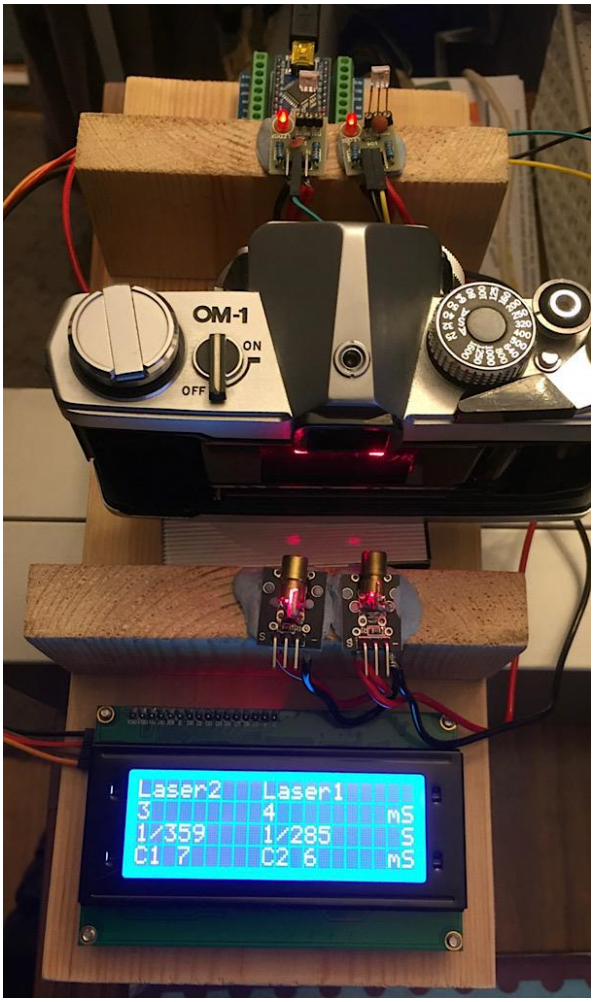
Chopping board used for base and electrical socket boxes used to house sensors & Lasers.

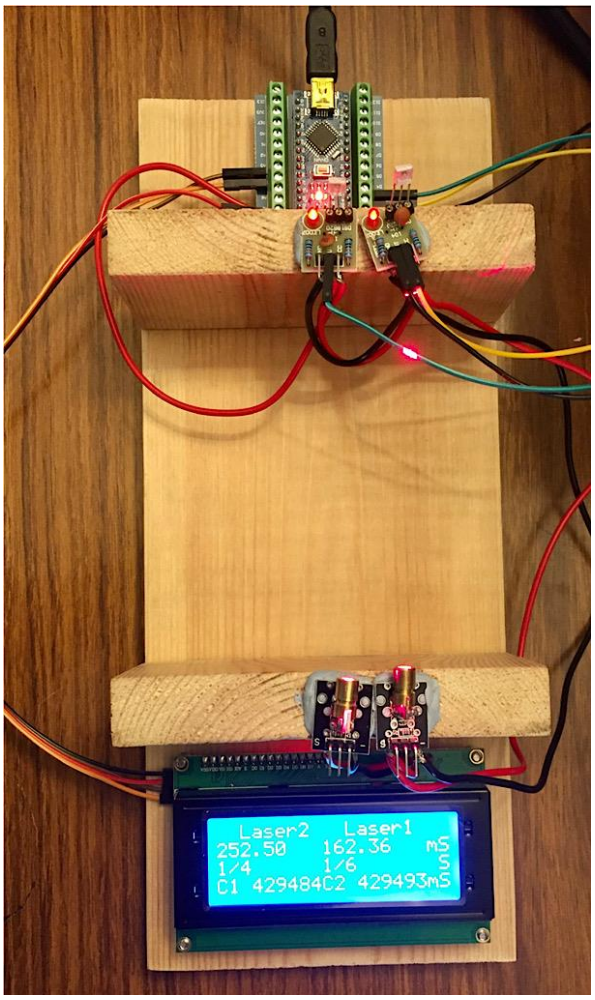


Mask added to the Lego version, for better accuracy at higher shutter speeds.



Simple ‘block of wood’ sensor frame.


OM-1
ON
OFF
Laser2 Laser1
3 4 mS
1/359 1/285 S
C1 7 C2 6 mS


Laser2 Laser1
252.50 162.36 mS
1/4 1/6 S
C1 429484C2 429493mS